# Lesson 7

## Installing and Configuring an OS

# Topic 7A

## Install an Operating System

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Exam Objectives Covered

- 2.1 Given a scenario, install server operating systems

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Installation Process

- OS source files
- Live installations
- The installer
- Manual versus unattended

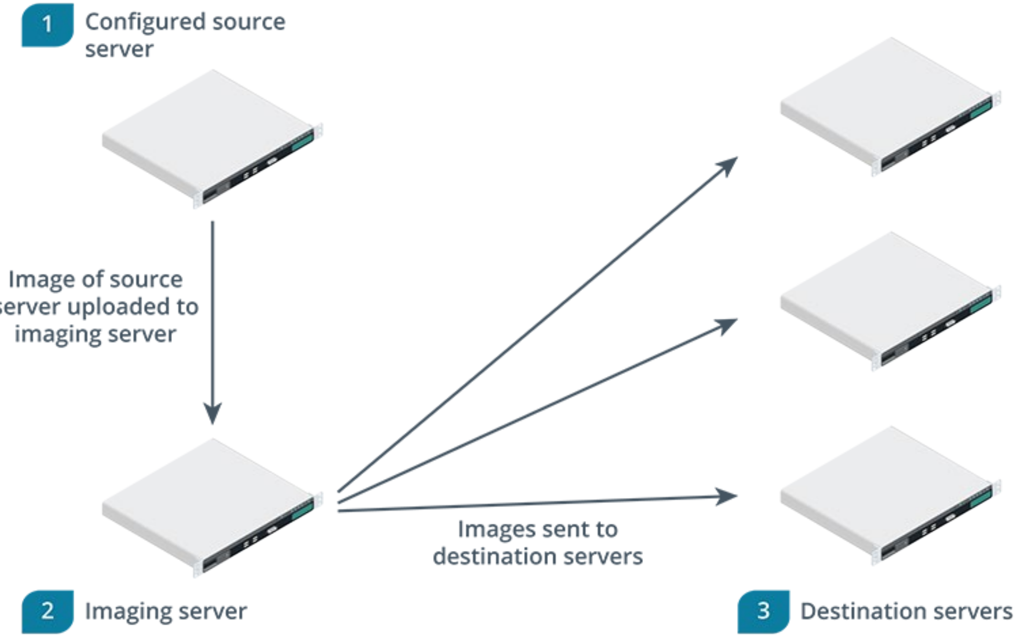
```
[root@localhost ~]# cat anaconda-ks.cfg
# Generated by Anaconda 33.25.4
# Generated by pykickstart v3.29
#version=F33
# Use graphical install
graphical


# Keyboard layouts
keyboard --xlayouts='us'
# System language
lang en_US.UTF-8

# Network information
network  --hostname=localhost.localdomain

# Run the Setup Agent on first boot
firstboot --enable

# Generated using Blivet version 3.3.0
ignoredisk --only-use=sda
autopart
# Partition clearing information
clearpart --none --initlabel

# System timezone
timezone America/Denver --utc

#Root password
rootpw --lock
```

*Linux Kickstart files are used for automated, unattended installations.*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Installation Process (slide 2)

- Imaging




*The source computer's configuration is uploaded to a deployment server. The image is then deployed to a destination server. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Installation Process (slide 3)

- Server migrations
- Bare metal versus virtualized installations

 Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. 

# Managing the Server with a GUI or CLI

- Linux

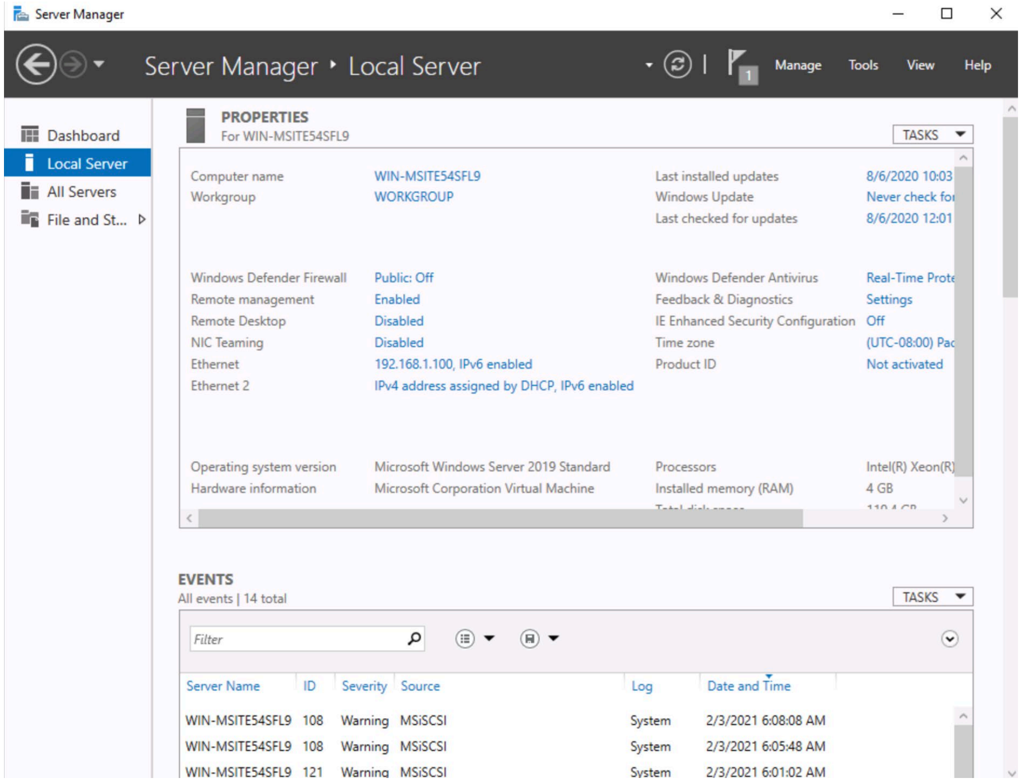
```
[root@localhost ~]# uptime
 07:58:31 up 54 min,  1 user,  load average: 0.00, 0.07, 0.04
[root@localhost ~]# hostname
localhost.localdomain
[root@localhost ~]# ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:12:67:76 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83120sec preferred_lft 83120sec
    inet6 fe80::4ba0:590f:a7ca:1827/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]# 
```

*Linux administrative commands.*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Managing the Server with a GUI or CLI (slide 2)

- Windows




*Windows Server Manager. (Screenshot courtesy of Microsoft.)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Managing the Server with a GUI or CLI (slide 3)

- Remote administration
  - SSH
  - RDP
  - VNC


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Guidelines for Installing an Operating System

- Hardware
- Automate
- Image deployments
- VM templates
- Cloud deployments


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review Activity: OS Installation

Review Activity

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Topic 7B

## Configure Storage

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Exam Objectives Covered

- 2.1 Given a scenario, install server operating systems


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Volumes and Partitions

- Windows
- Linux


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

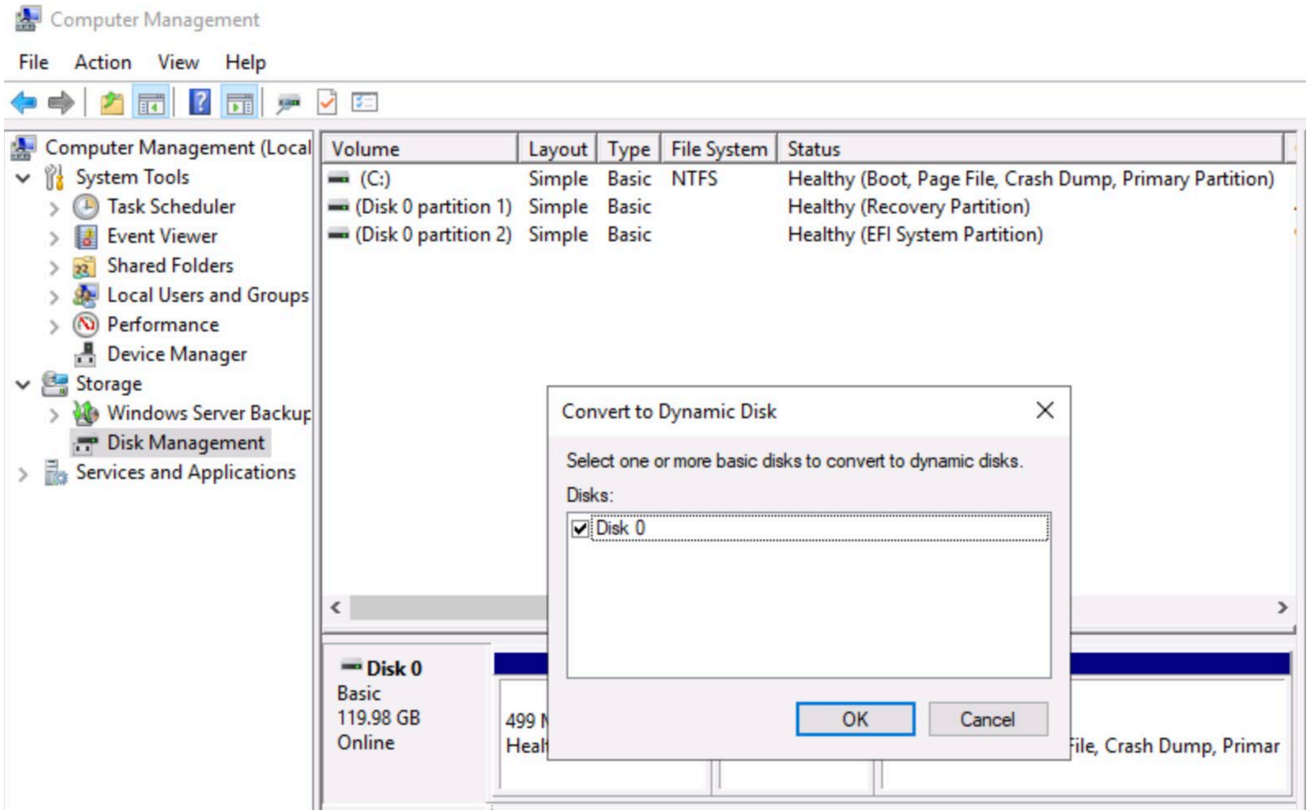
# Partition Tables

- Master Boot Record (MBR)
- Guid Partition Table (GPT)


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Partition and Volume Management

- Partitions
  - Primary
  - Extended
- Windows disks
  - Basic
  - Dynamic

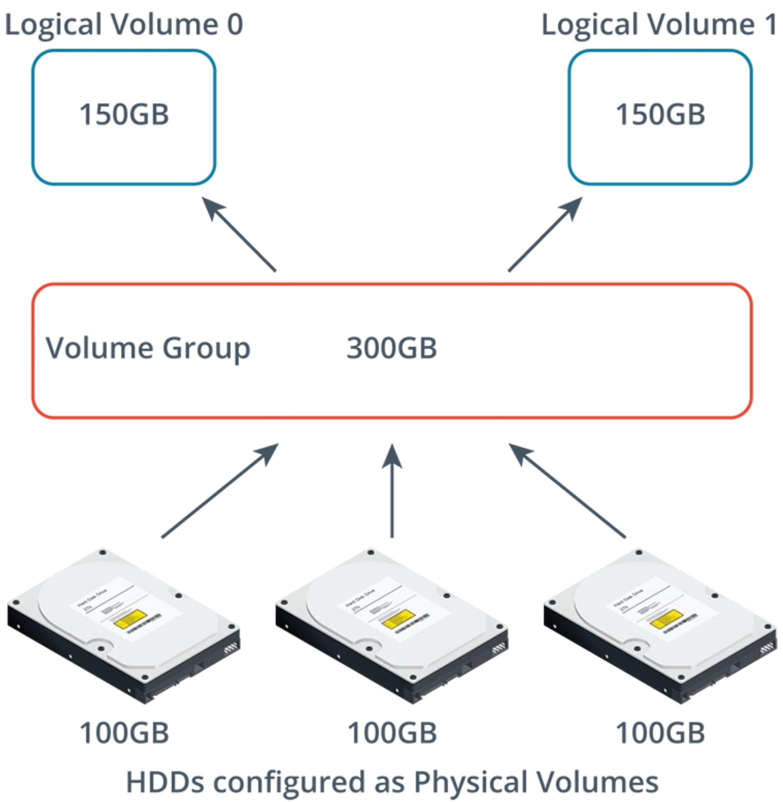



*The conversion option in the Windows Server Disk Management console. (Screenshot courtesy of Microsoft.)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Partition and Volume Management (slide 2)

- LVM in Linux
- PVs
- VGs
- LVs




*The three layers of LVM. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Partition Strategies

- Windows
- Linux
- Windows File Server
- Linux Web Server


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# File Systems

- ext4
- XFS

```
[root@CentOS7-A ~]# xfs_info /dev/sda3
meta-data=/dev/sda3              isize=512    agcount=4, agsize=1310720 blks
         =                       sectsz=4096  attr=2, projid32bit=1
         =                       crc=1        finobt=0 spinodes=0
data     =                       bsize=4096   blocks=5242880, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0 ftype=1
log      =internal               bsize=4096   blocks=2560, version=2
         =                       sectsz=4096  sunit=1 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@CentOS7-A ~]#
```

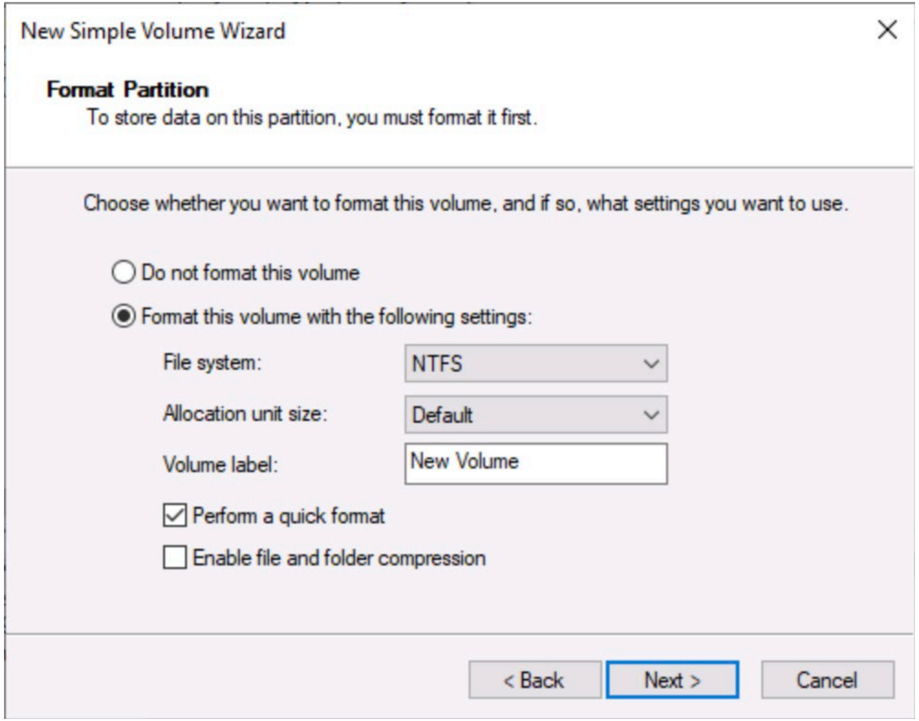
*XFS configuration information on a Linux server.*

- ZFS


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# File Systems (slide 2)

- NTFS
- ReFS
- VMFS




*Format a new volume with the NTFS file system. (Screenshot courtesy of Microsoft.)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Guidelines for Configuring Storage

- GPT for > 2TB
- Windows Server has basic and dynamic disk options and uses NTFS
- Linux leverages LVM and uses XFS and ext4
- Windows OS on C:\ and user data on D:\
- Linux uses root (/), home, and var

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review Activity: Storage Configuration

Review Activity

# Topic 7C

## Configure Network Settings

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Exam Objectives Covered

- 2.2 Given a scenario, configure servers to use network infrastructure services


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review Network Fundamentals

- OSI model
- TCP/IP suite

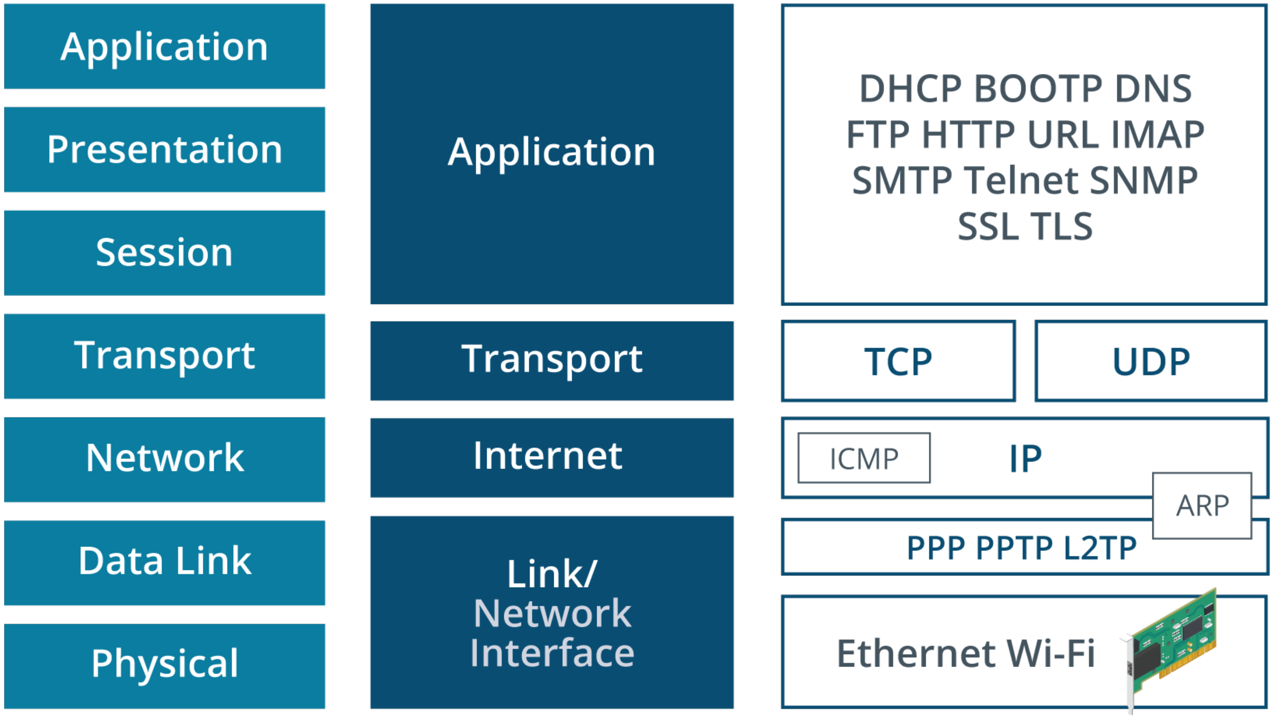

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review the OSI Model

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review the TCP/IP Suite




*The Application, Transport, Internet, and Network Access Layers of TCP/IP. (Images © 123RF.com)*

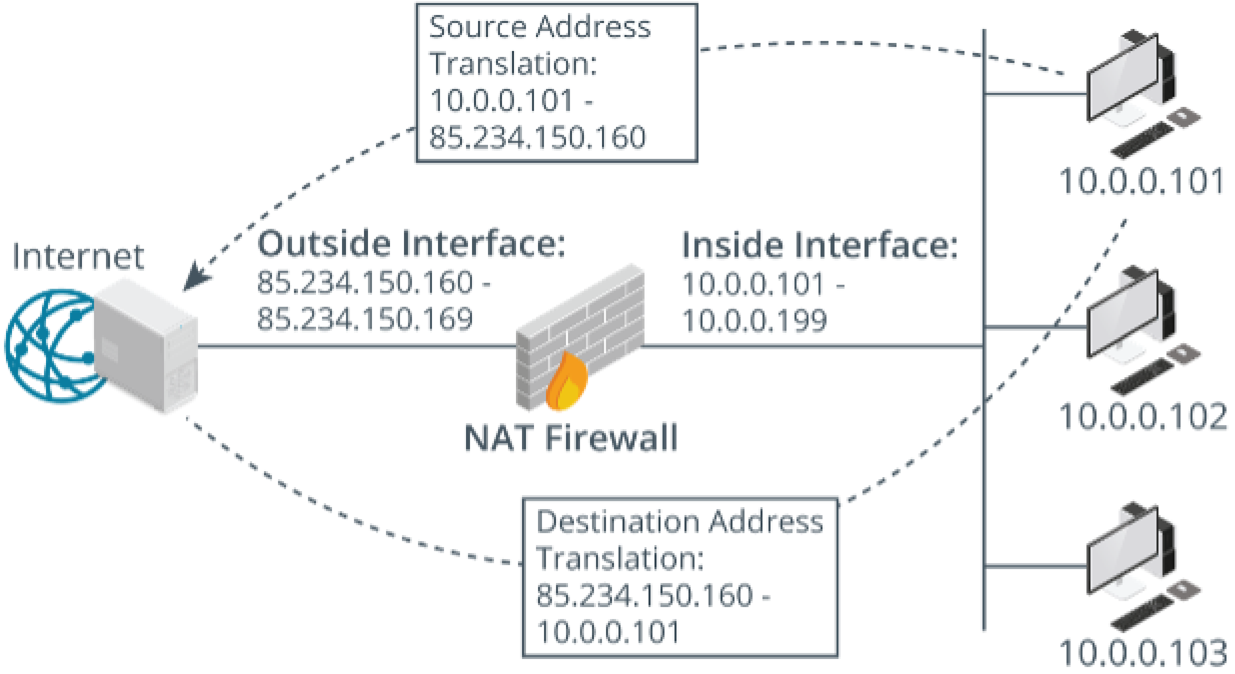

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Addressing

- IPv4 addresses
- The loopback and APIPA addresses
- Network ID, host ID, and the subnet mask
- Bit count notation

# Addressing (slide 2)

- Network address translation




*NAT device translating between public and private IP addresses. (Images © 123RF.com)*

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Addressing (slide 3)

- IPv4 address depletion
- IP address management
- IPv6 addresses

```
PS C:\Users\Administrator> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::217f:3a98:d74b:ca21%13
   IPv4 Address. . . . . . . . . . . : 192.168.1.100
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1:1%13
                                       192.168.1.1
```

*Output from the ipconfig command displaying the IPv6 address.*

# Network Identities

- MAC addresses
- IP addresses
- Hostnames

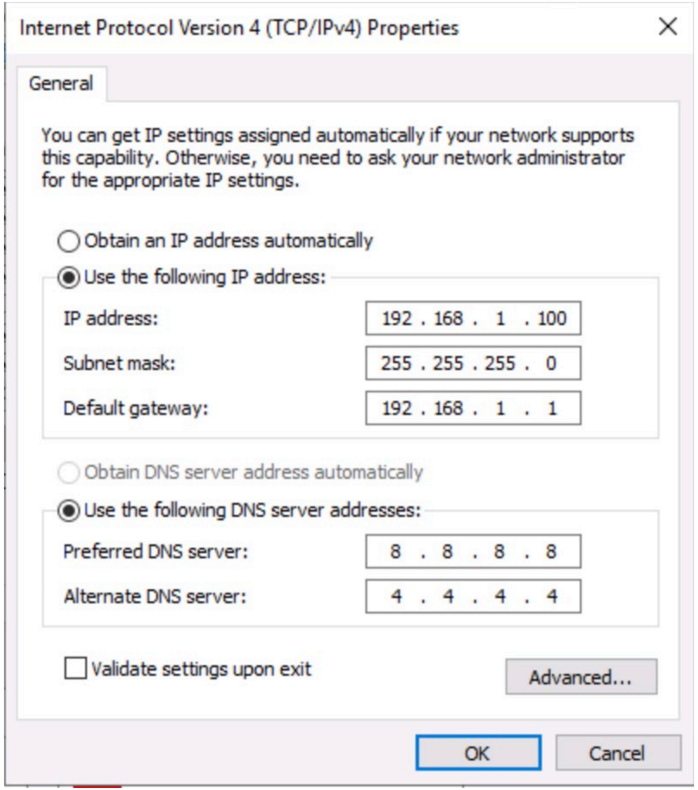
```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : WIN-MSITE54SFL9
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : localdomain

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : Microsoft Hyper-V Network Adapter
   Physical Address. . . . . . . . . : 00-15-5D-01-80-08
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::217f:3a98:d74b:ca21%13(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.100(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1:1%13
                                       192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 100668765
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-25-B2-71-FD-00-15-5D-01-80-08
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       4.4.4.4
   NetBIOS over Tcpip. . . . . . . . : Enabled
   Connection-specific DNS Suffix Search List :
                                       localdomain
```

*Ipconfig /all output, note the physical (MAC) address, the IPv4 Address, and the Host Name.*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# IP Address Configuration

- IP address configuration methods
- Static method



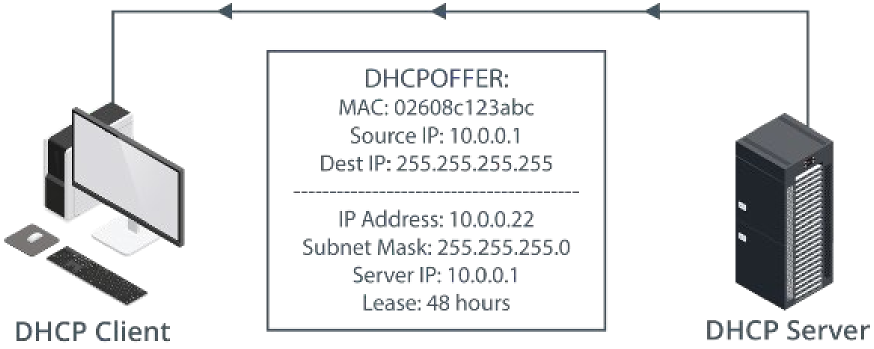

*The configuration dialog box to statically set an IP address in Windows. (Screenshot courtesy of Microsoft.)*

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# IP Address Configuration (slide 2)

- Dynamic method


*Step one of the four-step DHCP lease generation process: Discover (Images © 123RF.com)*

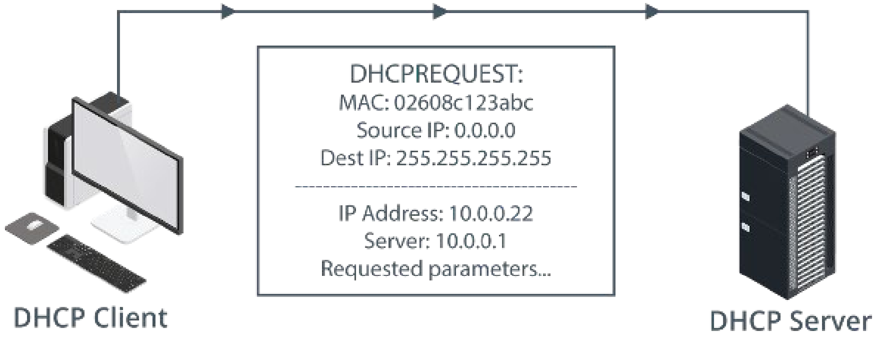



*Step two of the four-step DHCP lease generation process: Offer (Images © 123RF.com)*

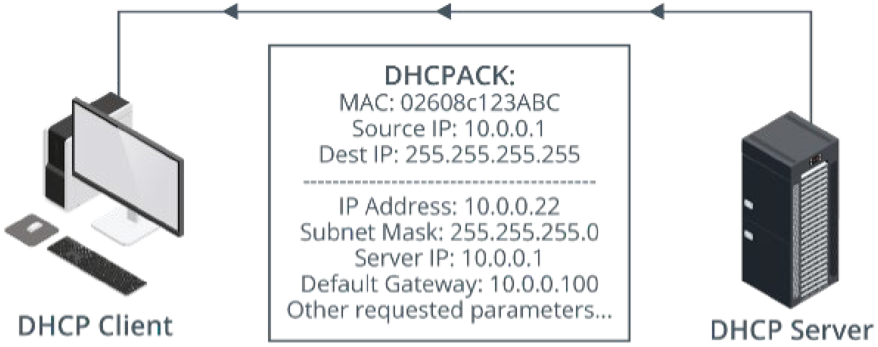

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# IP Address Configuration (slide 3)

- Dynamic method




*Step three of the four-step DHCP lease generation process: Request (Images © 123RF.com)*




*Step four of the four-step DHCP lease generation process: Acknowledge (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# IP Address Configuration (slide 4)

- DHCP lease
- APIPA address range (169.254.0.0)

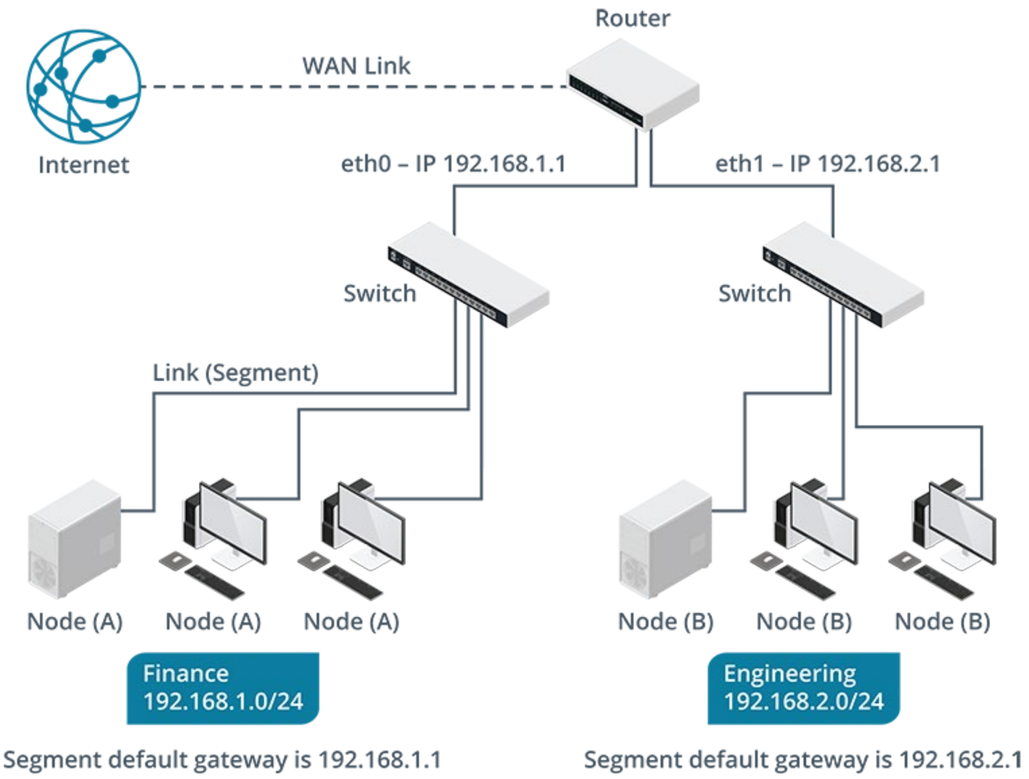
```
C:\Users\LabUser>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-BT4131D
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : localdomain

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : Microsoft Hyper-V Network Adapter
   Physical Address. . . . . . . . . : 00-15-5D-01-80-00
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::35a2:35c7:3721:90fe%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.101(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, February 4, 2021 7:20:31 PM
   Lease Expires . . . . . . . . . . : Thursday, February 4, 2021 9:20:31 PM
   Default Gateway . . . . . . . . . : fe80::1:1%14
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
```

*DHCP lease information displayed by using the ipconfig /all command.*

CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Network Segmentation

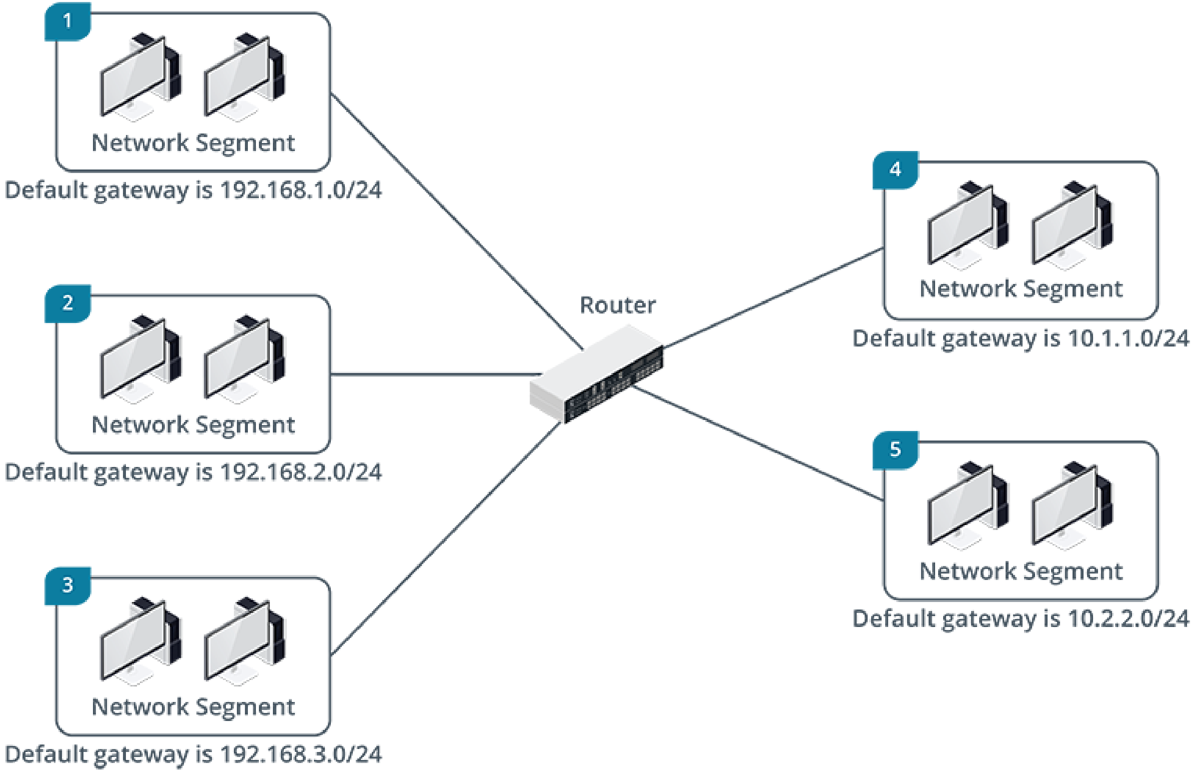
- Segmentation




*A simple network displaying the router's IP addresses for each NIC, the network IDs and default gateways. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Network Segmentation (slide 2)

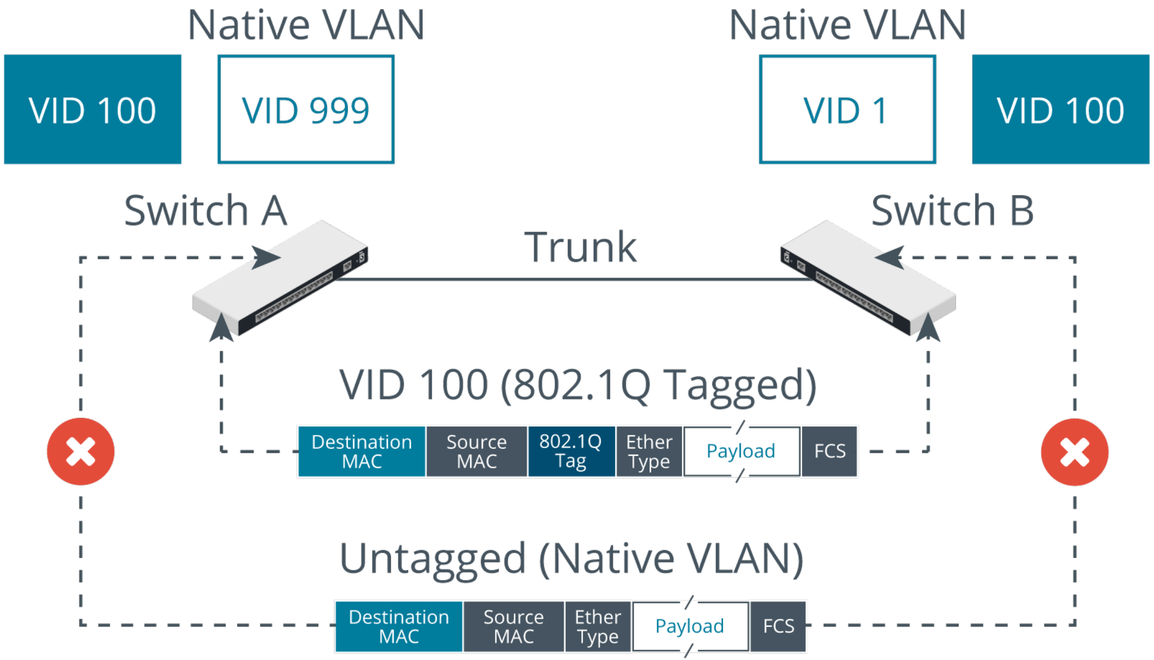



*A network with five segments and one router. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Network Segmentation (slide 3)

- VLAN




*A VLAN with multiple switches. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Name Resolution

```
[root@localhost ~]# host www.comptia.org
www.comptia.org is an alias for production-comptiawebsite.trafficmanager.net.
production-comptiawebsite.trafficmanager.net is an alias for production-northcentral-www.comptia.org.
production-northcentral-www.comptia.org has address 23.96.239.26
[root@localhost ~]# nslookup 52.165.6.154
** server can't find 154.6.165.52.in-addr.arpa: NXDOMAIN

[root@localhost ~]#
```

*A website URL resolved to an IP address.*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Name Resolution (slide 2)

- FQDN
- Hosts files

```
[root@localhost ~]# cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.2.200   workstation01 workstation01.localdomain         #User workstation
192.168.2.201   devworkstation devworkstation.localdomain       #Developer workstation
[root@localhost ~]#
```

*The /etc/hosts file.*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Name Resolution (slide 3)

- DNS

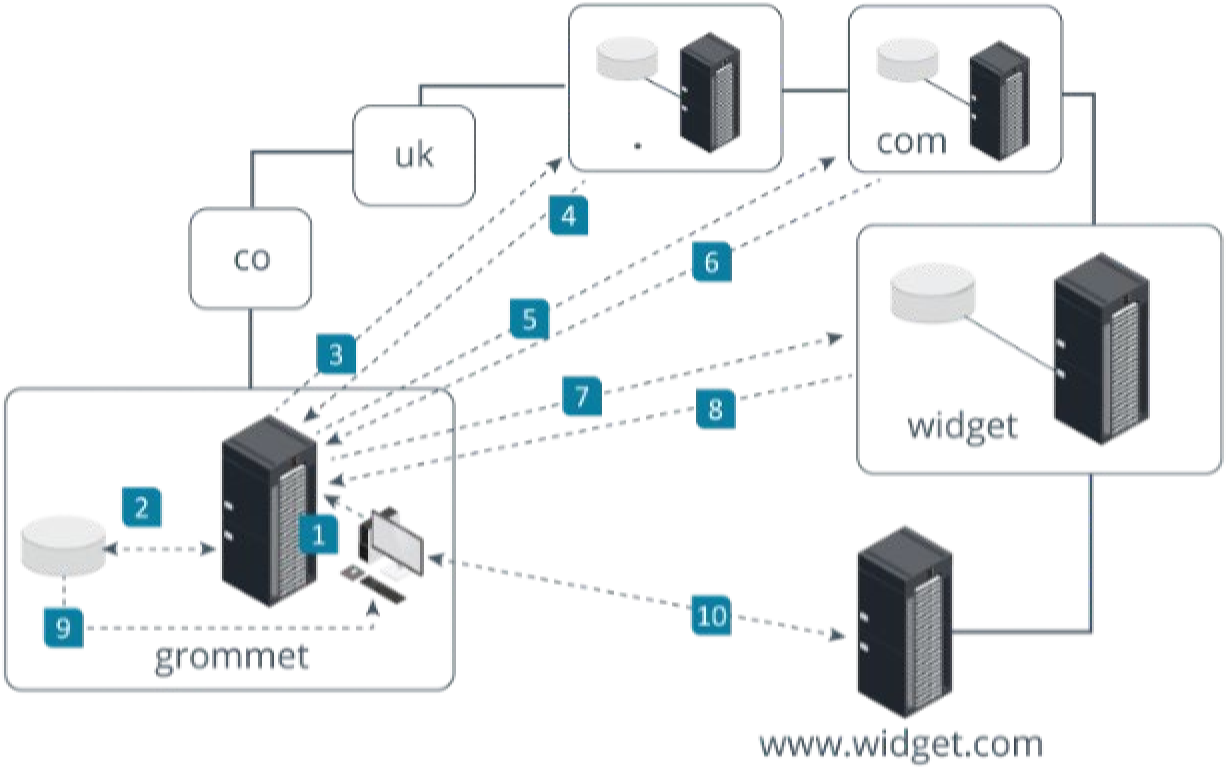
| | | | |
|---|---|---|---|
| (same as parent folder) | Host (A) | 192.168.0.102 | 30/12/2017 |
| 99servera | Host (A) | 192.168.0.188 | static |
| FileServer1 | Host (A) | 192.168.0.178 | static |
| MailServer1 | Mail Exchanger (MX) | [10] serverb99.widgets.co... | static |
| MailServer2 | Mail Exchanger (MX) | [10] servera99.widgets.co... | static |
| ServerA99 | Host (A) | 192.168.0.12 | static |
| serverb99 | Host (A) | 192.168.0.102 | static |
| ServerC99 | Host (A) | 192.168.0.34 | static |
| TestServer | IPv6 Host (AAAA) | 2001:0001:3dfe:0001:0000:... | static |
| www | Alias (CNAME) | serverc99.widgets.com. | static |

*The DNS records on a Windows Server DNS Server. (Screenshot courtesy of Microsoft.)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Name Resolution (slide 4)

- DNS




*Example of a client computer querying DNS to resolve the IP address for a destination server identified by name. (Images © 123RF.com)*


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Network Devices

- Switch
- Router
- WAP

- Firewalls and port numbers


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Guidelines for Configuring Network Settings

- OSI model
- TCP/IP stack
- Common port numbers
- A node has a MAC address, IP address, and hostname
- Network ID, host ID, subnet mask
- Reserved IP address ranges
- Loopback and IPIPA addresses
- DHCP lease: Discover, Offer, Request, Acknowledge (DORA)
- Default gateway
- VLAN
- Name resolution


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org

# Review Activity: Network Settings

Review Activity

Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved.

# Lesson 7 pt. 1

## Summary


CompTIA Server+ SK0-005 | Copyright © 2021 CompTIA Properties, LLC. All Rights Reserved. | CompTIA.org